

PROJECT DOCUMENTATION

Metro Ticket Generating System in ServiceNow

Date	21 September 2026
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Project Name	Metro Ticket Generating System in ServiceNow

1. Introduction

1.1 Project Overview

Project Title: Metro Ticket Generating System in ServiceNow

This project moves metro ticket booking onto ServiceNow. Passengers use a Service Catalog form to book a ticket; the system works out the fare, records the chosen payment option, shows a QR-based digital ticket, and saves the booking details automatically in a custom table.

1.2 Purpose

The aim is to cut down manual ticketing work, make fares more accurate, offer several digital payment choices, give passengers a QR ticket, and keep operational records consistent and easy to trace.

1.3 Intended Users

Passengers are the main users. Station Managers and the Metro Operations Team use the resulting ticket records, and IT Admins maintain the catalog item, scripts.

1.4 Project Milestones

Catalog Form Setup → Fare Logic Script → QR Code Integration → Station Details Mapping → UI Pages → UAT & Deployment

2. Project Overview

The main features delivered by the solution are listed below.

Feature	Description
Station master data	Metro Station's Details table holding the list of stations.
Booking form	"Book A Metro Ticket" Service Catalog item with Starting From and Going To station references.
Journey and passengers	Single or Return journey, and a choice of 1 to 4 passengers.
Fare calculation	Fare worked out automatically from the selected stations and journey type.
Payment options	UPI, Card, or Others, with the Enter Payment Mode field appearing only for Others.
Digital ticket	Metro QR Widget that displays the QR code for the booking.

3. Architecture

The solution has three tiers, all built on the ServiceNow platform.

Tier	What it does
User Experience	ServiceNow Service Portal and Service Catalog present the booking form and confirmation.
Application Logic	Catalog Client Script handles fare calculation and validation; Catalog UI Policy controls conditional fields.
Database	Custom tables <i>u_metro_station_details</i> (stations).

4. Setup Instructions

Prerequisites: a ServiceNow developer instance (PDI) where you can create tables, catalog items, scripts, widgets, flows, and ACLs.

Carry out the steps in this order:

1. Create the table **Metro Station's Details** (name *u_metro_station_details*) from System Definition → Tables.
2. Add a single field, **Station Name** (*u_station_name*), and enter the station records.
3. Create the **Book A Metro Ticket** catalog item with its nine variables.
4. Configure the Fare Client Script so the amount updates when stations or journey type change.
5. Configure the Catalog UI Policy that reveals Enter Payment Mode when Others is selected.
6. Create the **Metro QR Widget** (ID *metro_qr_widget*) in Service Portal → Widgets, and the QR generation Catalog Client Script.

5. Configuration Structure

Area	Configuration
Master Data	Metro Station's Details — <i>u_metro_station_details</i>
Catalog Item	Book A Metro Ticket
Variables	Starting From, Going To, Type of journey, No of Passengers, Amount for single journey, Amount including return, Mode of Payment, Enter payment mode, QR Generated
Client Script	Fare Client Script
UI Policy	Conditional visibility for the Others payment mode
Widget	Metro QR Widget
Automation	Metro Project Flow
Security	Default ACLs on the custom tables
Numbering	MTKT followed by 5 digits

6. Running the Application

1. Log in to the ServiceNow PDI and copy the instance URL.
2. Open a new tab, paste the URL, and add **/sp** at the end to reach the Service Portal.
3. Search for and open **Book A Metro Ticket**.
4. Choose Starting From and Going To, then the journey type and number of passengers.
5. Check the calculated fare.
6. Select UPI, Card, or Others (enter the payment mode if Others).
7. Submit the request to complete the booking and generate the QR ticket.

7. API / Integration Documentation

Integration	Purpose	Input	Output
QRServer API	Creates the QR image for the ticket	Encoded ticket URL / identifier	QR image URL
ServiceNow Flow Designer	Creates the database record from the catalog request	Requested Item and catalog variables	Metro Database record

8. Authentication & Authorization

Login and identity are handled by the ServiceNow platform. Authorization relies on the four default ACLs that ServiceNow creates automatically with each custom table; no additional restriction ACLs were written for this project. In practice, passengers can book tickets and recharge through the catalog but cannot open the database tables directly. Tighter role-based rules on sensitive fields can be added later.

9. User Interface

The main interface is the Service Portal booking form. It groups the station, journey, passenger, fare, payment, and QR-related fields on one page, and the QR code is displayed by the Metro QR Widget once the ticket is generated.

10. Testing

Testing covers positive and negative catalog scenarios, fare calculation, conditional payment fields, QR display, Flow Designer record creation, RITM mapping, and access control. Record the actual outcome of each case during UAT.

Test Case	Steps / Input	Expected Result	Result
Single journey fare	Pick two different stations, Single journey	Amount for single journey is filled in	Pass / Fail
Return journey fare	Pick two stations, Return journey	Amount including return is calculated	Pass / Fail
Same station	Choose the same station for start and destination	Selection is rejected or flagged	Pass / Fail
Payment: Others	Select Others	Enter Payment Mode appears and is mandatory	Pass / Fail
Payment: UPI / Card	Select UPI or Card	Enter Payment Mode stays hidden	Pass / Fail
QR display	Submit a valid booking	QR code shows in the Metro QR Widget	Pass / Fail

11. Screenshots / Demo Evidence

Figure 1: Catalog item / configuration (variables, client scripts, UI policy)

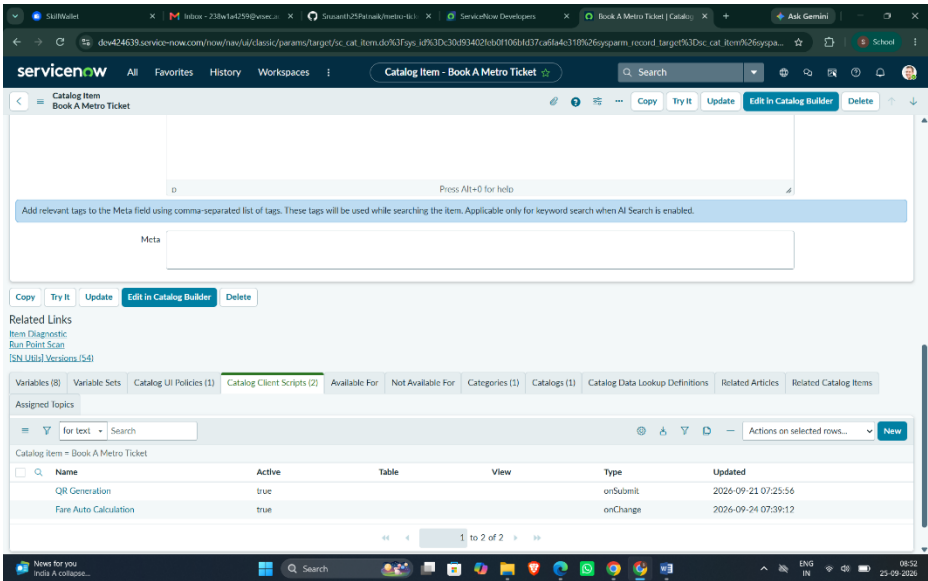


Figure 2: QR widget configuration

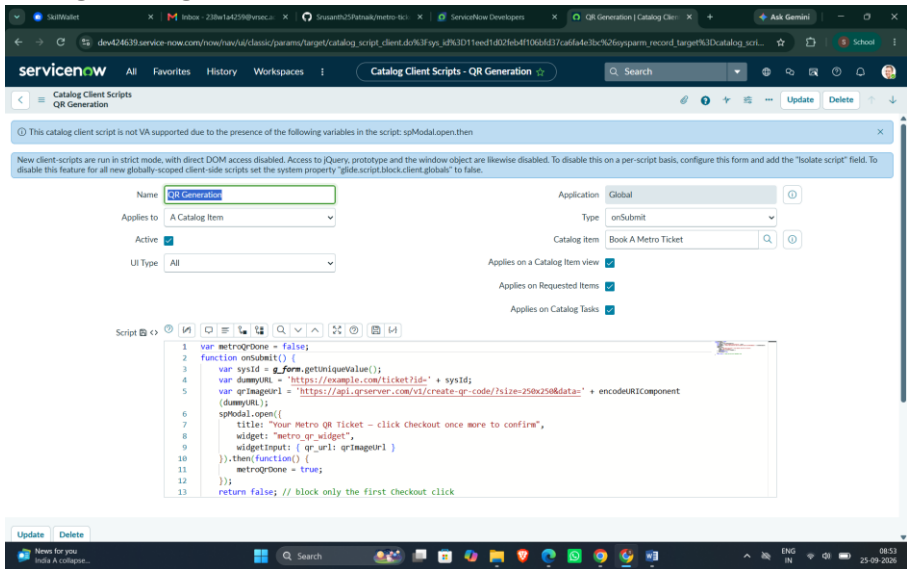


Figure 3: Metro ticket form on the Service Portal

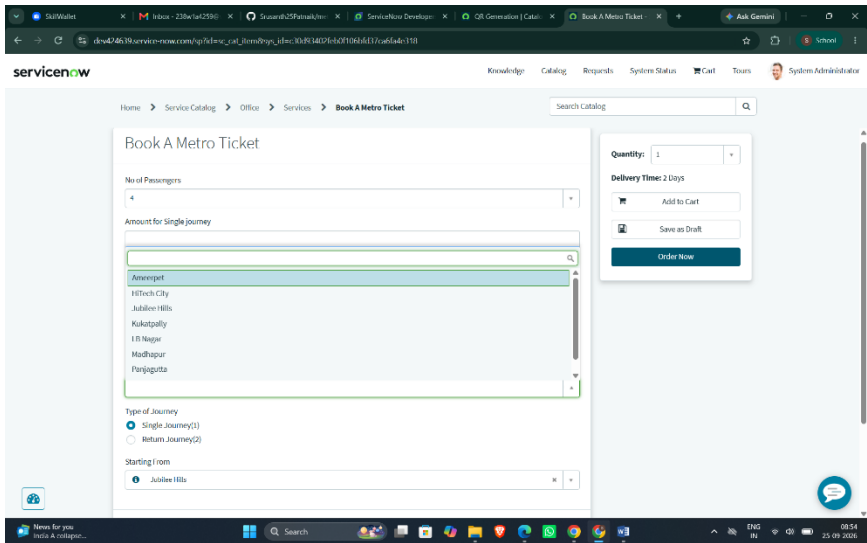


Figure 4: Completed booking with calculated fare

The screenshot shows the 'Book A Metro Ticket' form in the ServiceNow interface. The form is titled 'Book A Metro Ticket' and is located under the 'Service Catalog' > 'Office' > 'Services' > 'Book A Metro Ticket' path. The form fields include:

- No of Passengers:** A dropdown menu with the value '2' selected.
- Amount for Single journey:** A text input field with the value '40'.
- Mode of Payment:** Radio buttons for 'Upi' (selected), 'Card', and 'Others'.
- Amount including return:** A text input field.
- Going To:** A dropdown menu with the value 'Kakapally' selected.
- Type of Journey:** Radio buttons for 'Single Journey(1)' (selected) and 'Return Journey(2)'.
- Starting From:** A dropdown menu with the value 'Jubilee Hills' selected.

On the right side of the form, there is a sidebar with the following options:

- Quantity:** A dropdown menu with the value '1' selected.
- Delivery Time:** A text input field with the value '2 Days'.
- Add to Cart:** A button.
- Save as Draft:** A button.
- Order Now:** A button.

The ServiceNow logo is visible in the top left corner, and the user's name 'System Administrator' is in the top right corner. The browser address bar shows the URL 'dev424639.servicenow.com/sp?id=sc_cat_item&sys_id=c3d93402f6601064d37caf64c318'.

Figure 5: Metro QR

The screenshot shows a dialog box titled 'Your Metro QR Ticket — click Checkout once more to confirm'. The dialog box contains a QR code and the text 'Your Metro Ticket QR Code'. Below the QR code, it says 'Scan this QR code at the metro station gate'. The dialog box has 'Cancel' and 'OK' buttons. The background shows the 'Book A Metro Ticket' form from Figure 4, which is dimmed.

12. Known Issues

- **QR Generated** may stay false if the QR generation step is not triggered during a test.
- The Request Item field is deliberately kept as a reference; the readable RITM number is stored in a separate field.
- QR image generation depends on the external QR service being available.

13. Future Enhancements

- Integrate a real payment gateway.
- Validate QR codes at the metro gate in production.
- Replace the hard-coded fare matrix with a dynamic fare master table.
- Send passenger notifications by email or SMS.
- Add admin dashboards and usage analytics.
- Extend to more stations, zones, and fare rules.
- Optimize the ticket display for mobile.
- Promote changes through a controlled DEV → TEST → PROD update-set process.

Project Report Format

The final report follows the supplied report format as shown below.

Section	Covered By
1. Introduction	Sections 1 and 2 of this document
2. Ideation Phase	Ideation Phase document
3. Requirement Analysis	Requirement Analysis document
4. Project Design	Project Design Phase document
5. Project Planning & Scheduling	Project Planning Phase document
6. Functional and Performance Testing	Project Development Phase document and Section 10
7. Results	Demo screenshots and test outcomes
8. Advantages & Disadvantages	Table below
9. Conclusion	Conclusion below
10. Appendix	Configuration details, widget code

Advantages & Disadvantages

Advantages	Disadvantages / Limitations
Fare is calculated automatically	The fare matrix currently sits in client-side code
QR ticket is shown digitally	Relies on an external QR service
Ticket data is kept centrally in ServiceNow	No production payment integration
Several payment choices	Payment entry is informational, not a live gateway
Less manual effort	Full metro gate validation is outside the current scope

Conclusion

The Metro Ticket Generating System shows how ServiceNow can be set up as a digital metro ticketing workflow. A Service Catalog form is joined with automated fare logic, QR ticket display, Flow Designer automation, custom data tables, and request tracking, giving a complete path from booking to a stored ticket record.

Appendix

Item	Details
Key Table 1	u_metro_station_details — Metro Station's Details
Catalog Item	Book A Metro Ticket
QR Widget	Metro QR Widget (ID: metro_qr_widget)
Ticket Number	MTKT + 5 digits

Metro QR Widget — source code

HTML template:

```
<div style="text-align:center; padding:25px; font-family:Arial, sans-serif;">
  <h3>Your Metro Ticket QR Code</h3>
  
  <p>Scan this QR code at the metro station gate</p>
</div>
```

Server script:

```
(function() {
  data.qr_url = input ? input.qr_url : '';
})();
```